



Higher Technical School of Telecommunications Engineering and Systems



RADIOCOMMUNICATION SYSTEMS

PRACTICE 3

SOFTWARE-DEFINED RADIO (SDR) AND SINAD MEASUREMENT

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1. Introduction and Objectives

1.1. Introduction to Software-Defined Radio (SDR)

Definition

Software-defined radio (SDR) is a technology in which communication-system components traditionally implemented in hardware - such as mixers, filters, amplifiers and oscillators - are implemented in software. This means that the application or configuration of the system can be changed by modifying the software without altering the physical hardware.

General operation

- SDR devices can operate as receivers and transmitters.
- Multiple signals can be transmitted and received simultaneously.
- Receivers include an antenna, a signal-processing system and an interface for connection to the required device.

This practice uses the Analog Devices ADALM-PLUTO, an active-learning module that operates as a transceiver (transmitter and receiver).

- Transceptor con 1 transmisor y 1 receptor
- Rango de frecuencias útil: 325 MHz – 3,8 GHz
- Profundidad ADC y DAC: 12 bits
- Ganancia: 74,5 dB
- Salida USB
- Precio: 232,93 €

Figure from source page 1.



Figure from source page 1.

1.2. Objectives

The main objective is to become familiar with MATLAB and Simulink as tools for managing the complexity of an SDR communication system. The specific objectives are:

- Configure and control the ADALM-PLUTO from Simulink.
- Implement an FM broadcast receiver.
- Measure received-signal quality using SINAD (Signal-to-Noise and Distortion ratio).

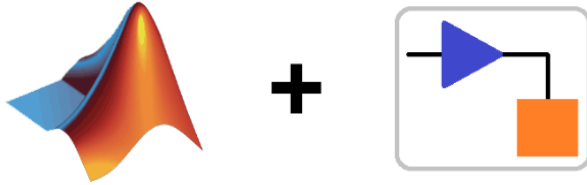


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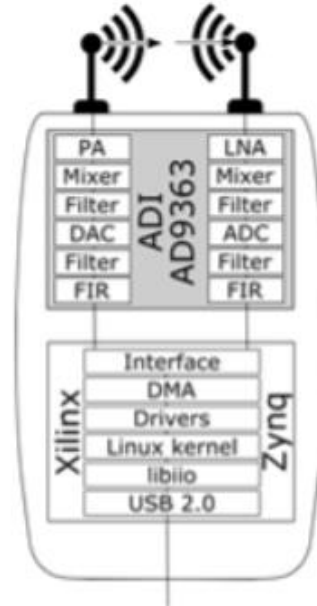


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2. Theoretical Basis: SINAD

SINAD (Signal-to-Noise and Distortion ratio) is a key parameter for evaluating the quality of a communication signal in the presence of disturbances. It is defined as the ratio between the total received power (signal + noise + distortion) and the disturbance power (noise + distortion).

When expressed in decibels and using RMS voltage values in Simulink, the ratio is calculated by subtracting logarithms: $\log(\text{numerator}) - \log(\text{denominator})$. The difference is multiplied by 20 to obtain SINAD in dB.

This parameter is important in telecommunications because it can be used to:

- Determine radio-receiver sensitivity, meaning the weakest signal that can still be decoded reliably.
- Evaluate audio-system fidelity and support radiocommunication-equipment maintenance by adjusting reception and sound quality.

$$SINAD = \frac{S + N + D}{N + D}$$

Figure from source page 3.

$$SINAD(dB) = 20 \cdot \log_{10} \left(\frac{V_{total}}{V_{ruido_distorsion}} \right)$$

Figure from source page 3.

$$SINAD(dB) = 20 \cdot (\log_{10}(RMS_{total}) - \log_{10}(RMS_{ruido}))$$

Figure from source page 3.

3. Experimental Procedure

3.1. Test Scenario

An RF generator and an antenna first transmit a 1 kHz sinusoidal signal, FM-modulated with a maximum frequency deviation of 75 kHz. The carrier is placed at a frequency not occupied by a commercial station, for example 100.2 MHz.

3.2. Simulink Receiver Design

The standard Simulink FM-receiver example was modified to include SINAD measurement. The final model contains the following blocks:

- 1 ADALM-PLUTO Receiver: configured to receive the signal at the selected center frequency.
- 2 FM Broadcast Demodulator: demodulates the RF signal and produces the baseband audio signal.
- 3 DC Blocker: placed after the demodulator to remove the DC component before measurement and prevent errors in the power calculation.
- 4 SINAD calculation: the circuit is divided into two parallel branches.
 - Numerator branch (complete signal): the signal passes directly to an RMS block, representing $S + N + D$.
 - Denominator branch (noise + distortion): a notch filter removes the useful 1 kHz tone. Its center frequency is 1000 Hz and its bandwidth is 200 Hz, leaving the $N + D$ residual.
- 1 Mathematical operation: the logarithms of both RMS values are calculated and subtracted; the result is multiplied by 20 and displayed in dB.

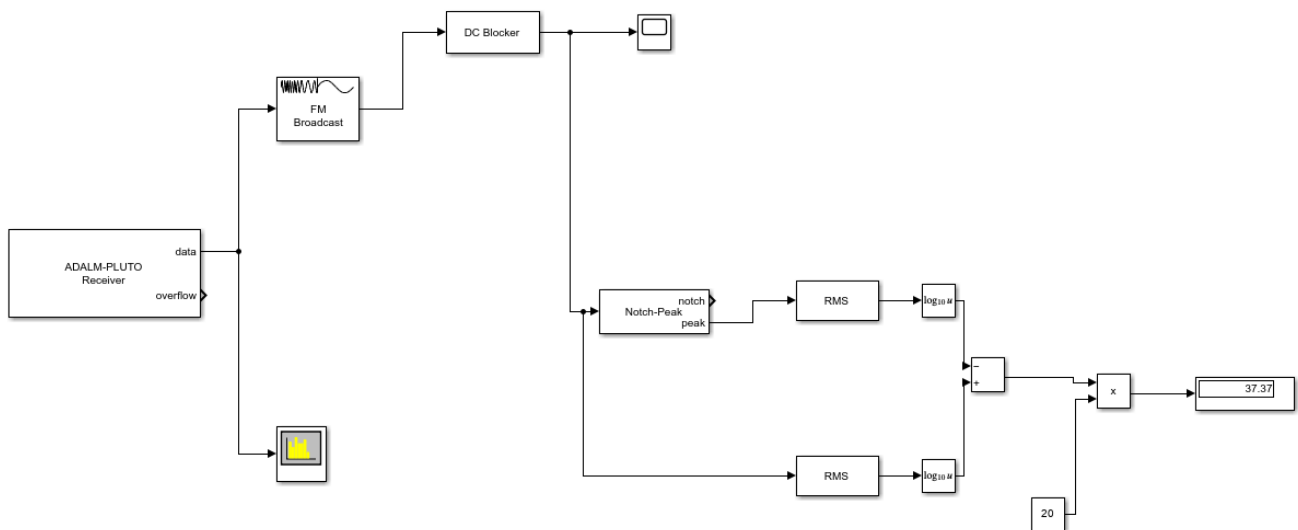


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4. Analysis of Results

4.1. Measurement 1 - Clean, Low-Power Signal

Frequency spectrum

The spectrum shows a centered FM signal with a peak power of approximately -46 dBm. The occupied bandwidth and the system noise floor are clearly visible.

Time-domain signal

The demodulated waveform is clean and sinusoidal, with no clipping or abrupt discontinuities. This indicates that the input stage was not saturated.



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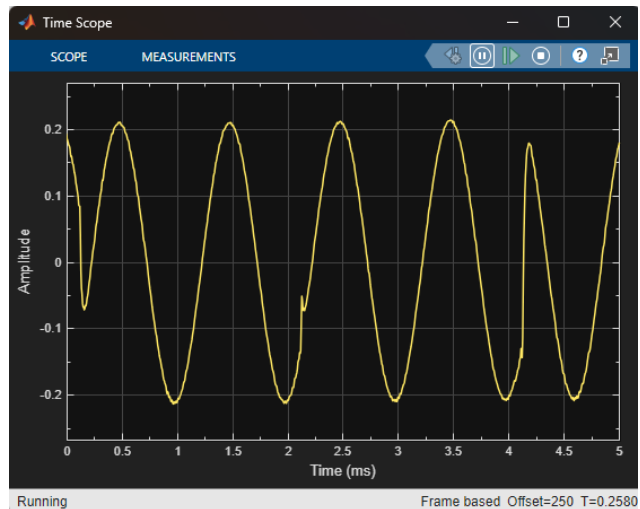


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SINAD measurement

Although the waveform is clean and distortion is low, the SINAD value is moderate. The useful-signal power is not especially high, so the margin above the thermal-noise level is limited.

4.2. Measurement 2 - Saturated, High-Power Signal

Frequency spectrum

The signal power increases substantially and the peak reaches approximately -34 dBm, about 12 dB above the first measurement.



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Time-domain signal

The higher input power produces visible distortion in the demodulated waveform, including small discontinuities near the sine-wave peaks. This indicates that a stage in the chain is approaching its dynamic-range limit or saturation.

SINAD measurement

Despite the visible distortion, this case produced the best SINAD result. The strong increase in useful-signal power more than compensates for the additional distortion. The signal is sufficiently strong relative to the background noise to achieve a higher final quality ratio than in the first case.

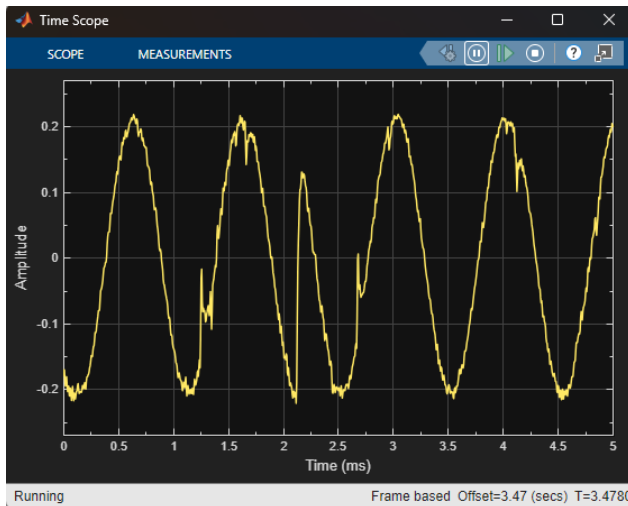


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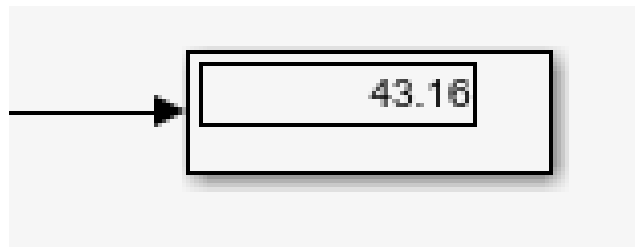


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5. Conclusions

This practice showed that SDR can provide a highly flexible platform for implementing, measuring and characterizing radio systems. With deeper study, the topic can become complex, but it is directly relevant to professional work in radio-system implementation and testing.

The main conclusions are:

- SDR technology: the ADALM-PLUTO and Simulink were used to implement a complete receiver and SINAD analyzer in software, without requiring an external hardware oscilloscope or spectrum analyzer.
- Measurement method: using a narrow 1 kHz notch filter with a 200 Hz bandwidth was an effective way to remove the fundamental audio tone and isolate the noise-and-distortion residual.
- Reception quality: the measured SINAD of 43.16 dB confirms that the laboratory communication link provides high-quality, low-noise audio transmission.
- Importance of the DC Blocker: removing the DC component is essential. Without this block, the RMS power measurement would include the DC offset and would bias the SINAD result downward.